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***B.Tech. Degree V Semester Special Supplementary Examination
April 2022***

**ME 15-1506 POWER PLANT ENGINEERING
(2015 Scheme)**

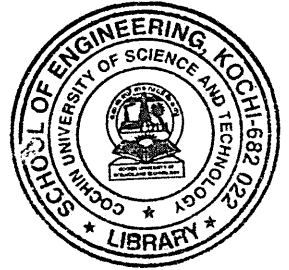
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Define Diversity factor and Load factor in respect of power plant economics.
- (b) What are the factors that affect the selection of a power plant Location?
- (c) What is the use of Surge Tank in Hydraulic dam and list the different types of surge tank?
- (d) What are the major component in a diesel Engine power plant?
- (e) Enumerate the advantage of gas turbine power plant over the diesel engine power plant.
- (f) Briefly explain Fluidized Bed Combustion (FBC).
- (g) Enumerate any six desirable characteristics for a reactor coolant.
- (h) What are the advantages and disadvantages of Geothermal Energy?
- (i) Explain the Principle of Fuel Cell.
- (j) Name different ash handling systems.



PART B

(4 × 10 = 40)

- II. (a) Name the elements that make up the operating cost of a power plant. (4)
- (b) A 60 MW power station has an annual peak load of 50MW. The power station supplies load having maximum demand of 20 MW, 17 MW, 10 MW and 9 MW. The annual load factor is 0.45. Find: (6)
 - (i) Average Load
 - (ii) Energy supplied per Year
 - (iii) Demand Factor.

OR

- III. Explain with a neat sketch, the essential features of a Hydro Electric power plant and different types of spill ways. (10)
- IV. (a) Name and briefly explain different types of fuel injection system in a diesel engine power plant. (5)
- (b) Explain the difference between wet sump lubrication system and dry sump lubrication system in a diesel engine. (5)

OR

- V. (a) Discuss different methods for the improvement of thermal efficiency of an open cycle Gas Turbine Power Plant. (7)
- (b) Enumerate the merits of gas turbine over IC engines. (3)

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- VI. (a) Enumerate the advantages and disadvantages of stoker firing. (4)
(b) Give a brief explanation on different types of dust collectors used in thermal power plants. (6)

OR

- VII. (a) What are the essential component of nuclear Reactor and give a brief explanation on Reactor Core. (5)
(b) Explain the parameters that affect the site selection of a nuclear power plant. (5)

- VIII. (a) With the help of neat sketch, explain solar pond Technology. (6)
(b) What are the advantages and disadvantages of Geothermal Energy? (4)

OR

- IX. (a) With the help of a neat sketch explain the working of a Janata Model Gobar Gas Plant. (7)
(b) Briefly explain the working of Thermionic Conversion System. (3)
